

MID TERM EXAMINATION

Paper: Mid Term Examination - Set A | Duration: 3 Hours | Max Marks: 20 | Date: 13-03-2026

Total Questions: 10 | Answer all questions.

Name: _____

Roll No: _____

Student Sign: _____

Teacher Sign: _____

SECTION A: MCQS

1. What is the primary intent of the Adapter pattern? [2 Marks]

- a) Decouple an abstraction from its implementation.
- b) Attach additional responsibilities to an object dynamically.
- c) Convert the interface of a class into another interface clients expect.
- d) Provide a unified interface to a set of interfaces in a subsystem.

2. Which design pattern aims to compose objects into tree structures to represent part-whole hierarchies, allowing clients to treat individual objects and compositions uniformly? [2 Marks]

- a) Decorator Pattern
- b) Composite Pattern
- c) Facade Pattern
- d) Bridge Pattern

3. According to the source text, what is a potential negative consequence of the Observer pattern? [2 Marks]

- a) It can complicate the design when not needed.
- b) It can introduce additional response time.
- c) It can cause memory leaks if observers aren't properly removed.
- d) It can become a god object coupled to all application classes.

4. A software system needs to apply various algorithms for data encryption, and these algorithms need to be easily swappable at runtime without altering the core logic that uses them. Which behavioral design pattern would be most appropriate for this scenario according to the provided text? [2 Marks]

- a) Command Pattern
- b) State Pattern
- c) Strategy Pattern
- d) Interpreter Pattern

5. In the context of the Flyweight pattern, what is the key distinction between intrinsic and extrinsic state? [2 Marks]

- a) Intrinsic state is mutable and context-dependent, while extrinsic state is shared and immutable.
- b) Intrinsic state is shared and immutable, while extrinsic state is context-dependent and passed to operations when needed.
- c) Intrinsic state is specific to each object instance, while extrinsic state is managed by the FlyweightFactory.
- d) Intrinsic state defines the interface of the Flyweight, while extrinsic state defines its implementation.

6. Consider a scenario where a complex library with numerous interdependent classes needs a simplified entry point for casual users, while still allowing advanced users direct access to the full functionality. Which structural design pattern is

[2 Marks]

7. A team is designing a system where multiple objects, referred to as 'colleagues,' need to interact with each other in complex, well-defined ways. To reduce direct dependencies between these colleagues and centralize their communication logic, which behavioral design pattern should they consider, and what is a risk associated with its implementation as per the text? [2 Marks]

- 8. Which structural design pattern focuses on providing a surrogate or placeholder for another object to control access to it, and what are two types of proxies explicitly mentioned in the source text? [2 Marks]**

9. Both the State pattern and the Strategy pattern involve a Context delegating behavior to separate objects (State or Strategy). Based on the provided text, describe a key conceptual difference in their intent and how they appear to the client. **[2 Marks]**

- 10.** The Decorator pattern and the Adapter pattern both involve 'wrapping' an existing class or component. Based on the provided text, what is a fundamental difference in their intent and how they interact with the client's expectations?**[2 Marks]**

- Decorator converts an interface for client compatibility, while Adapter adds responsibilities dynamically.
- Decorator's intent is to attach additional responsibilities dynamically, allowing flexibility over subclassing, whereas Adapter's
- Decorator focuses on composing objects into tree structures, while Adapter ensures efficient sharing of fine-grained objects.
- Decorator provides a simplified interface to a subsystem, while Adapter provides a surrogate for access control.

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